

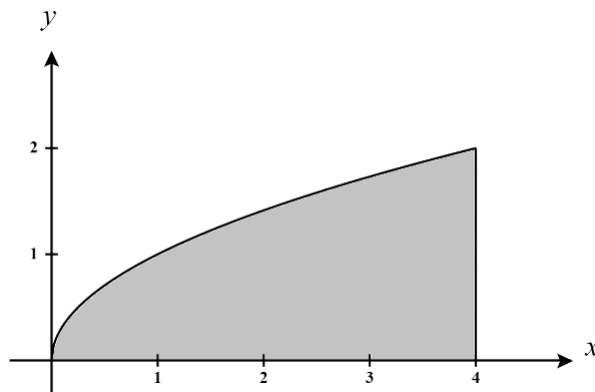
4.8 Centers of Gravity Using Multiple Integrals Worksheet

1. A lamina occupies the region $D = \{(x, y) \mid 0 \leq x \leq 4, 0 \leq y \leq \sqrt{x}\}$. Find the mass and center of mass of the lamina if its density at the point (x, y) is given by $\rho = 6xy$.

A picture of the region D is:

So, we note that the mass of the lamina is

$$\begin{aligned} m &= \iint_D 6xy \, dA = \int_0^4 \int_0^{\sqrt{x}} 6xy \, dy \, dx \\ &= \int_0^4 3xy^2 \Big|_0^{\sqrt{x}} \, dx = \int_0^4 3x^2 \, dx \\ &= x^3 \Big|_0^4 \\ &= 64 \end{aligned}$$



Then, by definition,

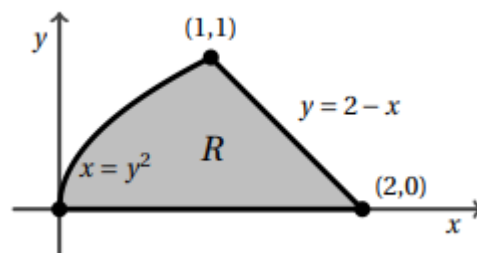
$$\begin{aligned} \bar{x} &= \frac{1}{64} \int_0^4 \int_0^{\sqrt{x}} (6x^2 y) \, dy \, dx = \frac{1}{64} \int_0^4 3x^2 y^2 \Big|_0^{\sqrt{x}} \, dx = \frac{1}{64} \int_0^4 3x^3 \, dx \\ &= \frac{1}{64} \left(\frac{3}{4} x^4 \right) \Big|_0^4 = \frac{1}{64} (3 \cdot 64) = 3 \end{aligned}$$

Then, by definition,

$$\begin{aligned} \bar{y} &= \frac{1}{64} \int_0^4 \int_0^{\sqrt{x}} (6xy^2) \, dy \, dx = \frac{1}{64} \int_0^4 2xy^3 \Big|_0^{\sqrt{x}} \, dx = \frac{1}{64} \int_0^4 2x(x^{3/2}) \, dx \\ &= \frac{1}{32} \int_0^4 x^{5/2} \, dx = \frac{1}{32} \left(\frac{2}{7} x^{7/2} \right) \Big|_0^4 \\ &= \frac{8}{7} \end{aligned}$$

Thus, the center of mass of the lamina is $\left(3, \frac{8}{7}\right)$.

2. Consider the region R pictured to the right. Determine the center of mass of this region if the region has a density determined by $\rho(x, y) = 4y$



We see that we can define R as

$$R = \{(x, y) \mid y^2 \leq x \leq 2 - y, 0 \leq y \leq 1\}$$

So, we note that the mass of the lamina is

$$\begin{aligned} m &= \iint_R \rho(x, y) dA = \int_0^1 \int_{y^2}^{2-y} (4y) dx dy \\ &= \int_0^1 (4xy) \Big|_{y^2}^{2-y} dy = \int_0^1 [4(2-y)y - 4y^3] dy \\ &= \int_0^1 (-4y^3 - 4y^2 + 8y) dy = \left(-y^4 - \frac{4}{3}y^3 + 4y^2 \right) \Big|_0^1 \\ &= \left(-1 - \frac{4}{3} + 4 \right) - (0) \\ &= \frac{5}{3} \end{aligned}$$

Then, by definition,

$$\begin{aligned} \bar{x} &= \frac{3}{5} \int_0^1 \int_{y^2}^{2-y} x(4y) dx dy \\ &= \frac{3}{5} \int_0^1 (2x^2 y) \Big|_{y^2}^{2-y} dy = \frac{3}{5} \int_0^1 [2(2-y)^2 y - 2y^5] dy = \frac{3}{5} \int_0^1 (8y - 8y^2 + 2y^3 - 2y^5) dy \\ &= \frac{3}{5} \left(4y^2 - \frac{8}{3}y^3 + \frac{1}{2}y^4 - \frac{1}{3}y^6 \right) \Big|_0^1 = \frac{3}{5} \left(4 - \frac{8}{3} + \frac{1}{2} - \frac{1}{3} \right) - (0) \\ &= \frac{9}{10} \end{aligned}$$

Then, by definition,

$$\begin{aligned} \bar{y} &= \frac{3}{5} \int_0^1 \int_{y^2}^{2-y} y(4y) dx dy \\ &= \frac{3}{5} \int_0^1 4y^2 x \Big|_{y^2}^{2-y} dy = \frac{3}{5} \int_0^1 [4y^2(2-y) - 4y^4] dy = \frac{3}{5} \int_0^1 (8y^2 - 4y^3 - 4y^4) dy \\ &= \frac{3}{5} \left(\frac{8}{3}y^3 - y^4 - \frac{4}{5}y^5 \right) \Big|_0^1 = \frac{3}{5} \left(\frac{8}{3} - 1 - \frac{4}{5} \right) - (0) \\ &= \frac{13}{25} \end{aligned}$$

Thus, the center of mass of the lamina is $\left(\frac{9}{10}, \frac{13}{25} \right)$.

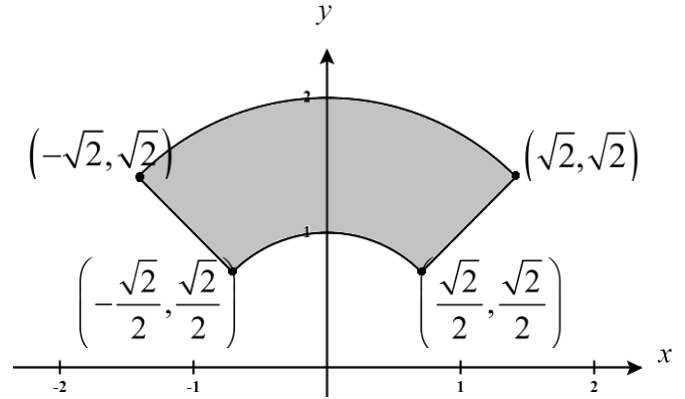
3. Consider the region R pictured to the right. Determine the center of mass of this region if the region has a density determined by $\rho(x, y) = x^2 + y^2$

We see that we can define R as

$$R = \left\{ (r, \theta) \mid 1 \leq r \leq 2, \frac{\pi}{4} \leq \theta \leq \frac{3\pi}{4} \right\}$$

So, we note that the mass of the lamina is

$$\begin{aligned} m &= \iint_R \rho(x, y) dA = \int_{\pi/4}^{3\pi/4} \int_1^2 (r^2) r dr d\theta \\ &= \int_{\pi/4}^{3\pi/4} \frac{1}{4} r^4 \Big|_1^2 d\theta = \int_{\pi/4}^{3\pi/4} \left(4 - \frac{1}{4} \right) d\theta \\ &= \int_{\pi/4}^{3\pi/4} \frac{15}{4} d\theta = \frac{15}{4} \left(\frac{\pi}{2} \right) \\ &= \frac{15\pi}{8} \end{aligned}$$



Then, by definition,

$$\begin{aligned} \bar{x} &= \frac{8}{15\pi} \int_{\pi/4}^{3\pi/4} \int_1^2 r \cos(\theta) (r^2) r dr d\theta = \frac{8}{15\pi} \int_{\pi/4}^{3\pi/4} \int_1^2 r^4 \cos(\theta) dr d\theta \\ &= \frac{8}{15\pi} \int_{\pi/4}^{3\pi/4} \frac{1}{5} r^5 \cos(\theta) \Big|_1^2 d\theta = \frac{8}{15\pi} \int_{\pi/4}^{3\pi/4} \left(\frac{32}{5} \cos(\theta) - \frac{1}{5} \cos(\theta) \right) d\theta = \frac{8}{15\pi} \int_{\pi/4}^{3\pi/4} \frac{31}{5} \cos(\theta) d\theta \\ &= \frac{248}{75\pi} (\sin(\theta)) \Big|_{\pi/4}^{3\pi/4} = \frac{8 \cdot 31}{75\pi} \left[\left(\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} \right) \right] \\ &= 0 \end{aligned}$$

Then, by definition,

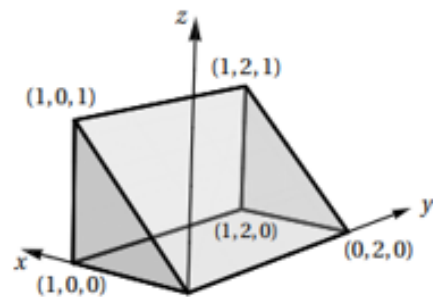
$$\begin{aligned} \bar{y} &= \frac{8}{15\pi} \int_{\pi/4}^{3\pi/4} \int_1^2 r \sin(\theta) (r^2) r dr d\theta = \frac{8}{15\pi} \int_{\pi/4}^{3\pi/4} \int_1^2 r^4 \sin(\theta) dr d\theta \\ &= \frac{8}{15\pi} \int_{\pi/4}^{3\pi/4} \frac{1}{5} r^5 \sin(\theta) \Big|_1^2 d\theta = \frac{8}{15\pi} \int_{\pi/4}^{3\pi/4} \left(\frac{32}{5} \sin(\theta) - \frac{1}{5} \sin(\theta) \right) d\theta = \frac{8}{15\pi} \int_{\pi/4}^{3\pi/4} \frac{31}{5} \sin(\theta) d\theta \\ &= \frac{248}{75\pi} (-\cos(\theta)) \Big|_{\pi/4}^{3\pi/4} = \frac{248}{75\pi} \left[\left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} \right) \right] \\ &= \frac{248\sqrt{2}}{75\pi} \end{aligned}$$

Thus, the center of mass of the lamina is $\left(0, \frac{248\sqrt{2}}{75\pi} \right)$.

4. Consider the solid region E shown to the right. Suppose the density at any point within E is given by $\rho(x, y, z) = z$. Determine the center of mass of the of E .

We first need to describe the given region. We see that the region is bounded below by 0 and above by the plane $z = x$. Therefore, we have $0 \leq z \leq x$.

If we project E into the xy -plane we produce a region D defined by $D = \{(x, y) | 0 \leq x \leq 1, 0 \leq y \leq 2\}$.



So, we note that the mass of the lamina is

$$\begin{aligned} m &= \iiint_E z \, dV = \int_0^1 \int_0^2 \int_0^x z \, dz \, dy \, dx = \int_0^1 \int_0^2 \frac{1}{2} z^2 \Big|_0^x \, dy \, dx \\ &= \int_0^1 \int_0^2 \frac{1}{2} x^2 \, dy \, dx = \int_0^1 \frac{1}{2} x^2 y \Big|_0^2 \, dx = \int_0^1 x^2 \, dx \\ &= \frac{1}{3} x^3 \Big|_0^1 \\ &= \frac{1}{3} \end{aligned}$$

By definition, we then find that:

$$\begin{aligned} \bar{x} &= 3 \iiint_E xz \, dV = 3 \int_0^1 \int_0^2 \int_0^x xz \, dz \, dy \, dx = \dots = \frac{3}{4} \\ \bar{y} &= 3 \iiint_E yz \, dV = 3 \int_0^1 \int_0^2 \int_0^x yz \, dz \, dy \, dx = \dots = 1 \\ \bar{z} &= 3 \iiint_E z^2 \, dV = 3 \int_0^1 \int_0^2 \int_0^x z^2 \, dz \, dy \, dx = \dots = \frac{1}{2} \end{aligned}$$

Thus, the center of mass of the solid is $\left(\frac{3}{4}, 1, \frac{1}{2}\right)$.

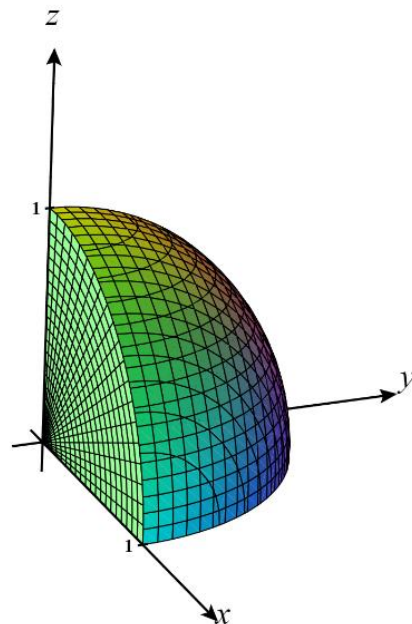
5. Let E be the portion of the positive octant which is inside the unit sphere. Determine the center of mass of the E assuming it has a density function $d(x, y, z) = \frac{1}{x^2 + y^2 + z^2}$.

We see that this region is defined by

$$E = \left\{ (\rho, \theta, \phi) \mid 0 \leq \rho \leq 1, 0 \leq \theta \leq \frac{\pi}{2}, 0 \leq \phi \leq \frac{\pi}{2} \right\}$$

So, we note that the mass of the lamina is

$$\begin{aligned} m &= \iiint_E \left(\frac{1}{x^2 + y^2 + z^2} \right) dV = \int_0^{\pi/2} \int_0^{\pi/2} \int_0^1 \left(\frac{1}{\rho^2} \right) \rho^2 \sin(\phi) d\rho d\theta d\phi \\ &= \int_0^{\pi/2} \int_0^{\pi/2} \int_0^1 \sin(\phi) d\rho d\theta d\phi = \int_0^{\pi/2} \int_0^{\pi/2} (\rho \sin(\phi)) \Big|_0^1 d\rho d\theta d\phi \\ &= \int_0^{\pi/2} \int_0^{\pi/2} \sin(\phi) d\theta d\phi = \int_0^{\pi/2} \frac{\pi}{2} \sin(\phi) d\phi \\ &= -\frac{\pi}{2} \cos(\phi) \Big|_0^{\pi/2} = -\frac{\pi}{2}(0) + \frac{\pi}{2}(1) \\ &= \frac{\pi}{2} \end{aligned}$$



By definition, we then find that:

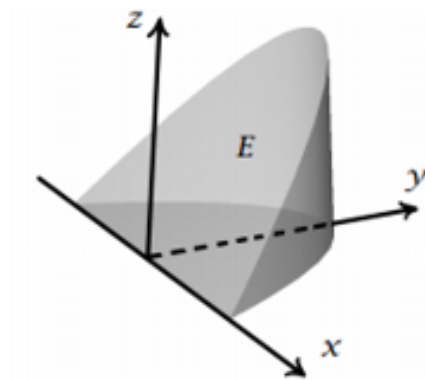
$$\begin{aligned} \bar{x} &= \frac{4}{\pi} \iiint_E x \left(\frac{1}{x^2 + y^2 + z^2} \right) dV = \int_0^{\pi/2} \int_0^{\pi/2} \int_0^1 (\rho \sin(\phi) \cos(\theta)) \left(\frac{1}{\rho} \right) \rho^2 \sin(\phi) d\rho d\theta d\phi \\ &= \int_0^{\pi/2} \int_0^{\pi/2} \int_0^1 \rho^2 \sin^2(\phi) \cos(\theta) d\rho d\theta d\phi = \dots = \frac{1}{3} \end{aligned}$$

$$\begin{aligned} \bar{y} &= \frac{4}{\pi} \iiint_E y \left(\frac{1}{x^2 + y^2 + z^2} \right) dV = \int_0^{\pi/2} \int_0^{\pi/2} \int_0^1 (\rho \sin(\phi) \sin(\theta)) \left(\frac{1}{\rho} \right) \rho^2 \sin(\phi) d\rho d\theta d\phi \\ &= \int_0^{\pi/2} \int_0^{\pi/2} \int_0^1 \rho^2 \sin^2(\phi) \sin(\theta) d\rho d\theta d\phi = \dots = \frac{1}{3} \end{aligned}$$

$$\begin{aligned} \bar{z} &= \frac{4}{\pi} \iiint_E z \left(\frac{1}{x^2 + y^2 + z^2} \right) dV = \int_0^{\pi/2} \int_0^{\pi/2} \int_0^1 (\rho \cos(\phi)) \left(\frac{1}{\rho} \right) \rho^2 \sin(\phi) d\rho d\theta d\phi \\ &= \int_0^{\pi/2} \int_0^{\pi/2} \int_0^1 \rho^2 \sin(\phi) \cos(\phi) d\rho d\theta d\phi = \dots = \frac{1}{3} \end{aligned}$$

Thus, the center of mass of the solid is $\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right)$.

6. Consider the region E shown below which is bounded the xy -plane, the plane $z - y = 0$ and the surface $x^2 + y = 1$. Determine the center of mass of this cone assuming it has a density function $\rho(x, y, z) = xz$.



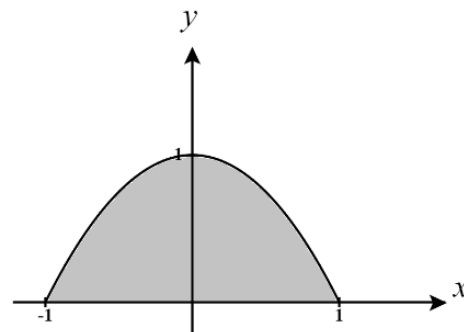
We first need to describe the region E . We see that the region is bounded below by 0 and above by the plane $z = y$. Therefore, we have $0 \leq z \leq y$.

If we project the solid into the xy -plane we produce a region D shown below and can be defined by

$$D = \{(x, y) \mid -1 \leq x \leq 1, 0 \leq y \leq 1 - x^2\}.$$

So, we note that the mass of the lamina is

$$\begin{aligned} m &= \int_{-1}^1 \int_0^{1-x^2} \int_0^y x^2 dz dy dx = \int_{-1}^1 \int_0^{1-x^2} x^2 y dy dx \\ &= \int_{-1}^1 \left. \frac{1}{2} x^2 y^2 \right|_0^{1-x^2} dx = \int_{-1}^1 \frac{1}{2} x^2 (1-x^2)^2 dx = \int_{-1}^1 \left(\frac{1}{2} x^2 - x^4 + \frac{1}{2} x^6 \right) dx \\ &= \left(\frac{1}{6} x^3 - \frac{1}{5} x^5 + \frac{1}{14} x^7 \right) \Big|_{-1}^1 = \left(\frac{1}{6} - \frac{1}{5} + \frac{1}{14} \right) - \left(-\frac{1}{6} + \frac{1}{5} - \frac{1}{14} \right) \\ &= \frac{8}{105} \end{aligned}$$



By definition, we then find that:

$$\bar{x} = \frac{105}{8} \iiint_E x \cdot x^2 dV = \frac{105}{8} \int_{-1}^1 \int_0^{1-x^2} \int_0^y x^3 dz dy dx = \dots = 0$$

$$\bar{y} = \frac{105}{8} \iiint_E y \cdot x^2 dV = \frac{105}{8} \int_{-1}^1 \int_0^{1-x^2} \int_0^y yx^2 dz dy dx = \dots = \frac{4}{9}$$

$$\bar{z} = \frac{105}{8} \iiint_E z \cdot x^2 dV = \frac{105}{8} \int_{-1}^1 \int_0^{1-x^2} \int_0^y zx^2 dz dy dx = \dots = \frac{2}{9}$$

Thus, the center of mass of the solid is $\left(0, \frac{4}{9}, \frac{2}{9} \right)$.